ECSE 318 - VLSI

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Questions 1 - 2

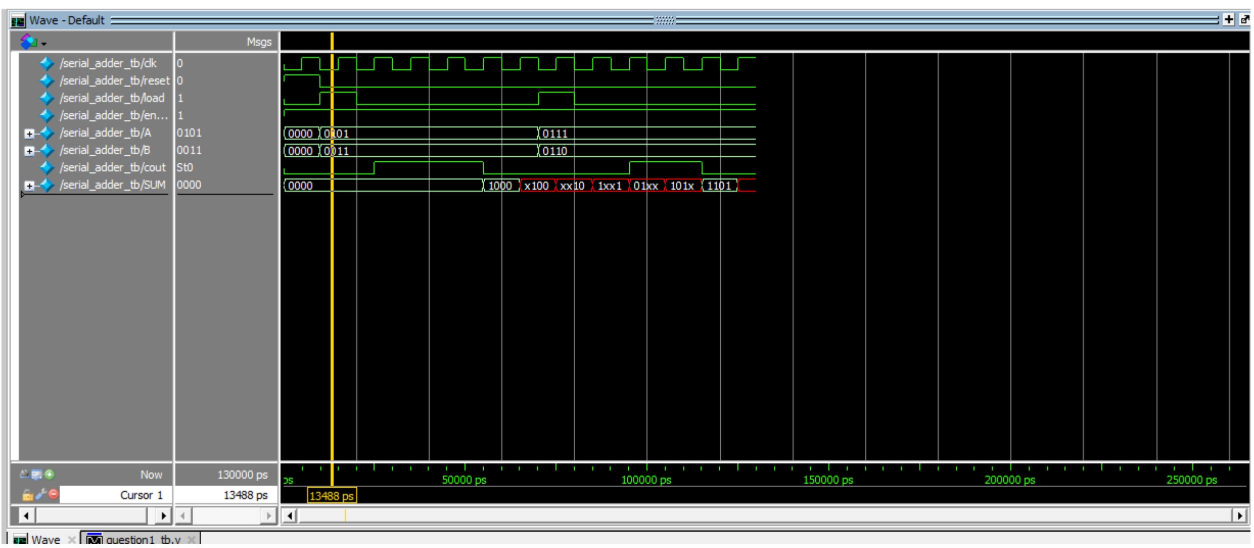
Question 1

a) Structural model

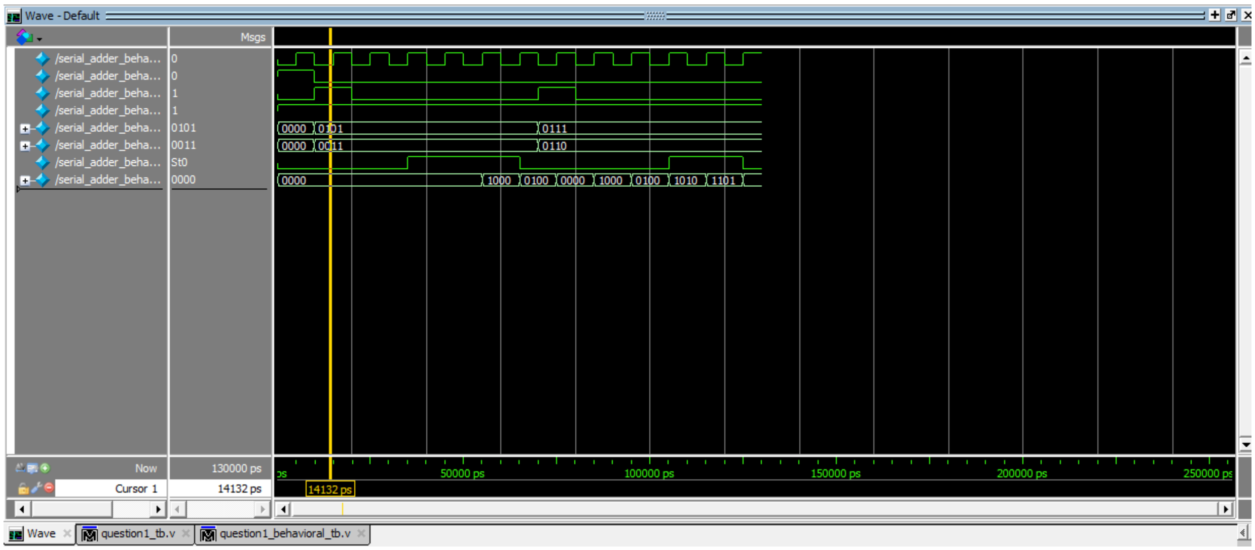
b) Behavioral model

c) The models are equivalent as the waveforms verify that both models exhibit the same functionality by yielding the same results when given the same inputs.

Pictured below: waveform for structural

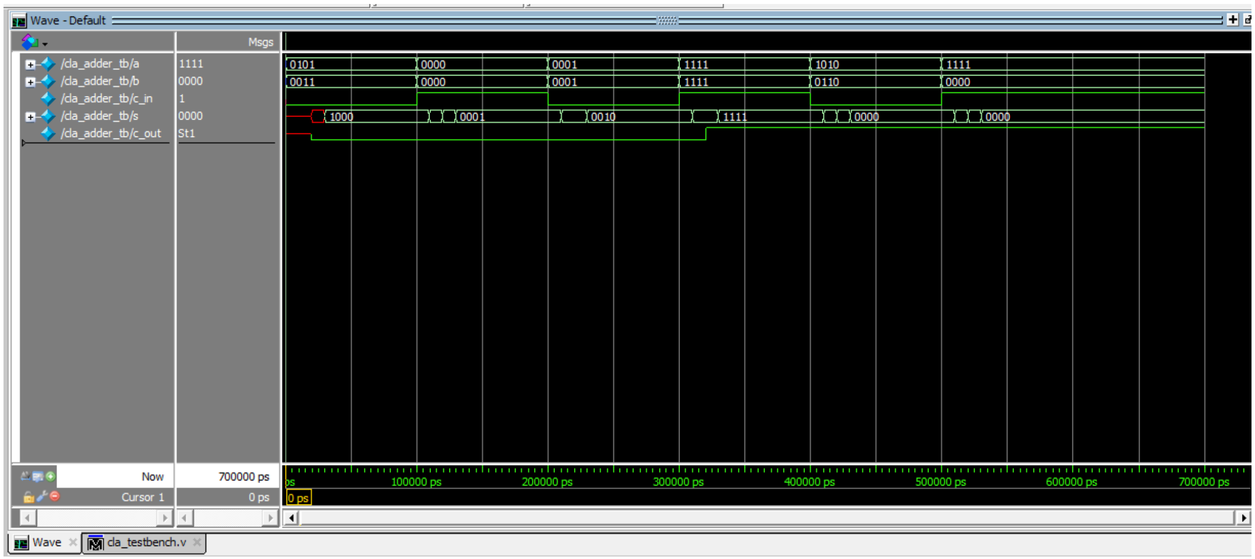


Pictured below: waveform for behavioral



Question 2

a) 4 bit CLA



b) Derive an input that exhibits the largest delay from C0 the MSB and find the actual delay:

   i) The input that has the largest delay is a = 1111, b = 0000, and c = 1. This is because the carry in must take the longest path to be able to derive the output.

   ii) The typical delay for the tests was about 10000ps (100ns) which was the time to reach the correct result. Compared to the 20000ps (200ns), which was the time for the critical path to be able to derive the appropriate sum and carry out.

c) Simulation output below

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# T=0 | a=0101 b=0011 c_in=0 | s=xxxx c_out=x
# T=2000 | a=0101 b=0011 c_in=0 | s=xxx0 c_out=0
# T=3000 | a=0101 b=0011 c_in=0 | s=1000 c_out=0
# T=10000 | a=0000 b=0000 c_in=1 | s=1000 c_out=0
# T=11000 | a=0000 b=0000 c_in=1 | s=1001 c_out=0
# T=12000 | a=0000 b=0000 c_in=1 | s=1111 c_out=0
# T=13000 | a=0000 b=0000 c_in=1 | s=0001 c_out=0
# At time                20000: Correct Result: Sum = 0001, Cout = 0
# T=20000 | a=0001 b=0001 c_in=0 | s=0001 c_out=0
# T=21000 | a=0001 b=0001 c_in=0 | s=0000 c_out=0
# T=23000 | a=0001 b=0001 c_in=0 | s=0010 c_out=0
# At time                30000: Correct Result: Sum = 0010, Cout = 0
# T=30000 | a=1111 b=1111 c_in=1 | s=0010 c_out=0
# T=31000 | a=1111 b=1111 c_in=1 | s=0011 c_out=0
# T=32000 | a=1111 b=1111 c_in=1 | s=0011 c_out=1
# T=33000 | a=1111 b=1111 c_in=1 | s=1111 c_out=1
# At time                40000: Correct Result: Sum = 1111, Cout = 1
# T=40000 | a=1010 b=0110 c_in=0 | s=1111 c_out=1
# T=41000 | a=1010 b=0110 c_in=0 | s=1110 c_out=1
# T=42000 | a=1010 b=0110 c_in=0 | s=0010 c_out=1
# T=43000 | a=1010 b=0110 c_in=0 | s=0000 c_out=1
# At time                50000: Correct Result: Sum = 0000, Cout = 1
# T=50000 | a=1111 b=0000 c_in=1 | s=0000 c_out=1
# T=51000 | a=1111 b=0000 c_in=1 | s=0001 c_out=1
# T=52000 | a=1111 b=0000 c_in=1 | s=0010 c_out=1
# T=53000 | a=1111 b=0000 c_in=1 | s=0000 c_out=1
# At time                70000: Correct Result: Sum = 0000, Cout = 1
```